

Text Analytics: Topic Modeling, Vocabulary Structure, and Document Embeddings

This section examines the latent semantic structure of the Active Inference corpus through complementary text-analytic methods: non-negative matrix factorization for topic discovery, TF-IDF vocabulary analysis, document embedding projections, and term co-occurrence patterns. Together, these analyses reveal thematic structure that cuts across the keyword-based domain taxonomy presented in the field overview.

Topic Modeling: Latent Structure

Non-negative matrix factorization (NMF) applied to the TF-IDF matrix identifies five latent topics:

Table 1: Non-negative matrix factorization (NMF) topic decomposition of the corpus TF-IDF matrix ($k = 8$ topics). Top terms are ranked by NMF component weight; interpretations reflect dominant thematic content.

Topic	Top Terms	Interpretation
0	learning, agent, model, agents, active, environments, aif, inference, environment, based	Agent-environment model
1	inference, active, energy, free, variational, control, bayesian, expected, optimal, principle	Active inference agents
2	states, internal, external, systems, markov, system, dynamics, information, beliefs, self	Markov blankets and internal states
3	fep, systems, ai, principle, energy, free, theory, networks, modeling, language	Free energy principle and modeling
4	predictive, brain, cognitive, prediction, perception, processing, sensory, models, coding, model	Predictive coding and sensory processing